Statistics for Data Science

Assignment 3

Choosing Statistical Tests and Building Regression Models

Group Assignment

Submission deadline: *2026.11.25 2:30pm*

Group size: A maximum of six students.

Submission Instructions

1. Submit one PDF report and one reproducible `.R` script per group.
2. At the beginning of the PDF report, before your answers, include a **Division of Work** section listing every group member's name, student ID, and a brief contribution statement. Follow the format shown below. No separate contribution file is required.
3. Organise the report by question. Include relevant tables, labelled plots, and interpretations in your own words. Do not submit an unedited collection of console outputs or screenshots.
4. Unless otherwise stated, use $\alpha = 0.05$ and 95% confidence intervals.
5. R may be used for all numerical work. No manual calculation of test statistics is required. Report results to three decimal places where appropriate. Report $p < 0.001$, not $p = 0.000$.
6. Explain how missing values, invalid entries, and exclusions are handled. Do not fabricate observations or replace invalid entries with guessed values.
7. Acknowledge external sources, borrowed code, and any permitted AI assistance according to the course policy.

Division of Work: Required Format

Place the following information at the beginning of your PDF report. Replace the example group number, names, student IDs, and contributions with your group's actual details.

```
Alice Chen --- 12345678 --- Data preparation and variable classification
Bob Li --- 12345679 --- Question 1 and Question 2
Carol Wang --- 12345680 --- Predictor reasoning
David Zhao --- 12345681 --- Model selection and interpretation
Eve Sun --- 12345682 --- Question 3
```

Every group member must be listed. Members not listed will be treated as not having contributed.

Assessment Overview

Question	Topic	Points
1	Advertising responses: choosing a test	10
2	Cars and fuel economy: comparing three means	20
3	Seven studies: selecting statistical tests	30
4	QQMusic: predicting song streams	40
	Total	**100**

General expectation:

Identify the research question, understand the variables and study design, choose an appropriate method, justify the choice, carry out the analysis in R, and interpret the evidence in context.

Part A: Statistical Tests

(60 points)

Question 1: Choosing an Advertisement (10 points)

A karate studio wants to compare three advertisements: Red, White, and Blue.

Different randomly selected consumers view each advertisement. Each consumer views only one advertisement and provides an ordered reaction score. A larger score indicates a more favourable response.

Use the supplied scores below.

```
red <- c(
  80, 80, 78, 81, 72, 85, 96, 84, 71, 75, 98
)

white <- c(
  75, 55, 98, 92, 86, 78, 87,
  79, 88, 87, 85, 94, 99
)

blue <- c(
  72, 76, 70, 77, 68, 82, 85, 81, 65, 69
)

ads <- data.frame(
  score = c(red, white, blue),
  advertisement = factor(
    rep(
      c("Red", "White", "Blue"),
      times = c(length(red), length(white), length(blue))
    )
  )
)
```

Question:

Is there evidence that consumer responses differ across the three advertisements?

(a) Identify the response scale, the number of groups, and whether observations are independent or paired. Select an appropriate test, explain your choice, and state the hypotheses. **[4 points]**

(b) Use R to summarise and plot the data, then carry out the selected test. Report the test statistic, degrees of freedom where applicable, p-value, and contextual conclusion. **[4 points]**

(c) Recommend an advertisement using the descriptive and inferential results. Explain what the overall test can and cannot establish about individual advertisements. No post-hoc tests are required. **[2 points]**

Question 2: Which Feed Helps Chicks Grow? (15 points)

The R dataset `chickwts` contains measurements from an experiment comparing six feed supplements. Newly hatched chicks were randomly allocated to six groups, and each group received a different feed supplement. The dataset records each chick's weight after six weeks.

There are 71 observations and two variables:

Variable	Description
`weight`	Body weight after six weeks, in grams
`feed`	Feed supplement: six categories

Load the data using:

```
data("chickwts", package = "datasets")
chicks <- chickwts
```

For this analysis, treat individual chicks as independent experimental units.

Question: Do the six feed types have the same population mean chick weight after six weeks?

(a) **Hypotheses and assumptions.** **[6 points]**

Define the population parameters and state the null and alternative hypotheses.

State the three main ANOVA assumptions and explain briefly how each can be assessed using the study description and the data. Use appropriate diagnostic evidence (words or graphs) to comment on the assumptions.

(b) **ANOVA in R.** **[7 points]**

Provide a labelled plot of weight by feed type. Fit a one-way ANOVA in R and create an ANOVA table showing source, degrees of freedom, sum of squares, mean square, F-statistic, and p-value. Report the statistical decision and answer the research question.

(c) **Interpretation.** **[2 points]**

A researcher states:

> "A significant ANOVA means that every feed type differs significantly from every other feed type. The feed with the highest sample mean is therefore significantly better than all the others."

Explain why this claim is not justified. Write a defensible conclusion from your analysis and explain what additional analysis would be needed to identify particular pairs that differ.

You are not required to conduct post-hoc tests.

Data source: R `datasets` package. See `help("chickwts")` for the original dataset description.

Question 3: Seven Studies: Which Statistical Test Would You Use? (30 points)

Use the real datasets supplied with `ISLR2` to analyse the seven studies below. For **each study**, provide a concise answer covering the following:

1. **Identify the design.**

   Describe the response scale or variable types, the number of groups or conditions, or whether observations are independent or paired where applicable.

2. **Choose and justify a test.**

   Select a method taught in the course. state the hypotheses.

3. **Analyse and interpret.**

   Work on the test in R, and report the statistic where applicable, degrees of freedom where applicable, p-value, decision, and contextual conclusion.

These are separate research questions. No adjustment across the seven studies is required.

Study A: Is There a League Salary Gap? (4 points)

The `Hitters` dataset contains Major League Baseball information from the 1986 and 1987 seasons.

Use `League`, which records the player's league at the end of 1986:

- `A`: American League;
- `N`: National League.

The variable `Salary` records the 1987 annual salary in thousands of US dollars.

Use players with an observed, positive salary and analyse the natural logarithm of salary.

```
hitters <- as.data.frame(ISLR2::Hitters)

hitters_salary <- subset(
  hitters,
  !is.na(Salary) & Salary > 0
)

hitters_salary$log_salary <- log(hitters_salary$Salary)
```

Question:

Do players in the two leagues have different population mean log salaries?

Study B: Tuition Versus Living Costs (4 points)

The `College` dataset contains information about 777 US colleges from the 1995 issue of *US News and World Report*.

Use:

- `Outstate`: out-of-state tuition;

- `Room.Board`: room and board costs.

Both are recorded in US dollars. Each college contributes both measurements.

```
colleges <- as.data.frame(ISLR2::College)

college_costs <- subset(
  colleges,
  !is.na(Outstate) &
    !is.na(Room.Board) &
    Outstate > 0 &
    Room.Board > 0
)

college_costs$log_tuition <- log(college_costs$Outstate)

college_costs$log_living <- log(college_costs$Room.Board)
```

Question:

Across the colleges represented, is the population mean difference between log out-of-state tuition and log room-and-board costs equal to zero?

Study C: Which Bill Is More Often Larger? (4 points)

Use the same `college_costs` data as Study B, but now focus only on which cost is larger at each college.

Question:

Among colleges where the two costs differ, is the probability that out-of-state tuition exceeds room-and-board costs equal to one half? Use a test without assuming normality or symmetry of their distribution. Compare the meaning of this result with the result from Study B.

Study D: American Versus Japanese Fuel Economy (4 points)

Use the `Auto` dataset and retain only American and Japanese cars.

```
cars_two <- subset(
  as.data.frame(ISLR2::Auto),
  origin %in% c(1, 3)
)

cars_two$origin_group <- factor(
  cars_two$origin,
  levels = c(1, 3),
  labels = c("American", "Japanese")
)
```

Question:

Is there evidence that the distributions of `mpg` differ between American and Japanese cars?

Study E: Are Public and Private Colleges Equally Represented? (4 points)

The variable `Private` in `College` indicates whether a college is private:

- `Yes`: private;
- `No`: public.

```
private_counts <- table(colleges$Private)
private_counts
```

Question:

Are the observed public and private college frequencies consistent with a model assigning probability 0.5 to each category? The equal-probability model is a hypothesis to test, not an established fact about US colleges.

Study F: Private Colleges and Incoming Student Profiles (5 points)

The variable `Top10perc` is the percentage of new students who came from the top 10% of their high-school class.

Define a college as having a `High` share when `Top10perc` is at least 50. Otherwise, classify it as `Lower`. This threshold is specified before the analysis. It is not a complete measure of college quality.

```
colleges$top10_group <- factor(
  ifelse(
    colleges$Top10perc >= 50,
    "High",
    "Lower"
  ),
  levels = c("Lower", "High")
)

private_top10 <- table(
  colleges$Private,
  colleges$top10_group
)

private_top10
```

Question:

Is private-college status associated with having a high share of incoming students from the top 10% of their high-school class? Explain how this research question differs from Study E and why an association would not demonstrate a causal effect of private-college status.

Study G: Which Origin Has More Variable Fuel Economy? (5 points)

Use the three origin groups from Question 2.

Question:

Is the population variance of `mpg` the same for American, European, and Japanese cars? Use a method taught in the course that compares variability across more than two groups without requiring normality.

R Notes for Question 3

- Levene's test is available through `car::leveneTest()`. State whether the test is centred on group means or group medians.
- Different players, colleges, or vehicles are treated as independent units for these analyses.

ISLR2 Data Summary

Dataset	Rows	Description
`Auto`	392	Automobile fuel economy and vehicle characteristics. Model years range from 1970 to 1982. Origin is coded as American, European, or Japanese.
`Hitters`	322	Major League Baseball player records from the 1986 and 1987 seasons. Salary contains missing values.
`College`	777	US college characteristics from the 1995 issue of *US News and World Report*. Includes college type, costs, and student profiles.

Consult the original package documentation:

```
help("Auto", package = "ISLR2")
help("Hitters", package = "ISLR2")
help("College", package = "ISLR2")
```

Mini Project

(40 points)

Question 4: What Makes a Hit? Exploring QQ Music (40 points)

Background and Scenario

Imagine that you are a data analyst working with the QQ Music team. The team wants to understand how song characteristics relate to music popularity and streaming performance.

The supplied dataset includes stream counts, release information, artist counts, platform indicators, and audio features. You will explore the data, develop your own research question, and answer it using a suitable generalized linear model (GLM).

Data Access

Download `QQ_Music.csv` from Moodle.

Questions

1. **Load the data into R, review the variables using the Data Dictionary below, and develop a research question through exploratory data analysis. [8 points]**

   (a) Explore the distributions of relevant variables and their relationships using EDA techniques introduced in this course. Include at least five informative plots, relevant summary statistics, and a brief discussion of your main findings.

   (b) Based on your EDA, propose one clear research question that can be addressed using a GLM. Identify your response variable and candidate predictors, and explain the question's relevance to the QQ Music team.

   Your response may be an original variable, a justified transformation, or a derived binary outcome. Justify any transformation or threshold used.

2. **For each variable considered in your research question, identify its: [5 points]**

   (a) data type or measurement scale (e.g., nominal, ordinal, interval, or ratio);

   (b) role as an independent or dependent variable;

   (c) role as a predictor or response variable.

   Assign analytical roles in relation to your research question. An identifier may be neither a predictor nor a response.

3. **Develop and evaluate a GLM to answer your research question. [27 points]**

   Use `glm()` with an appropriate family and link function. A Gaussian linear model fitted using `lm()` is also acceptable.

(a) **Predictor reasoning.** **[6 points]**
Justify the inclusion or exclusion of each candidate predictor using the research context and your EDA findings.

(b) **Correlation and multicollinearity.** **[4 points]**
Examine correlations among numerical predictors, excluding the response variable. Assess potential multicollinearity using appropriate diagnostics and explain how your findings inform the modelling decisions.

(c) **Model specification and selection.** **[7 points]**
Develop and compare candidate models using appropriate techniques. Justify your choice of response scale, family, link function, and final model.
Report the final model formulas and interpret two coefficients in context and provide a 95% confidence interval for at least two coefficient.

(d) **Diagnostics and prediction.** **[6 points]**
Evaluate your final model using diagnostic plots, statistics, and any additional checks appropriate to the chosen model.
Define a plausible response within the observed predictor ranges. Report its predicted outcome and an appropriate 95% uncertainty interval, explaining what the prediction and interval represent.

(e) **Conclusion and model justification.** **[4 points]**
Write approximately 200–300 words answering your research question. Justify your preferred model, summarise the main findings, and discuss the limitations of your analysis. Avoid interpreting associations as causal effects.

Data Description

Unit of observation: One song record.

Data Dictionary

Variable	Description
`track_name`	Name of the song.
`artist(s)_name`	Name(s) of the artist(s) contributing to the song.
`artist_count`	Number of artists contributing to the song.
`released_year`	Year in which the song was released.
`released_month`	Month in which the song was released.
`released_day`	Day of the month on which the song was released.
`in_QQ_playlists`	Number of QQ music playlists that include the song.
`in_QQ_charts`	Indicator of the song's presence and rank on QQ music charts.
`streams`	Total number of streams on QQ music recorded in the dataset.
`in_apple_playlists`	Number of Apple Music playlists that include the song.

Continued on next page

Variable	Description
`in_apple_charts`	Indicator of the song's presence and rank on Apple Music charts.
`bpm`	Beats per minute, representing the song's tempo.
`key`	Musical key of the song.
`mode`	Musical mode of the song: major or minor.
`danceability_%`	Score indicating how suitable the song is for dancing, expressed on a 0–100 scale.
`valence_%`	Score indicating the positivity of the song's musical content, expressed on a 0–100 scale.
`energy_%`	Score indicating the song's perceived energy level, expressed on a 0–100 scale.
`acousticness_%`	Score indicating the amount of acoustic sound in the song, expressed on a 0–100 scale.
`instrumentalness_%`	Score indicating the amount of instrumental content in the song, expressed on a 0–100 scale.
`liveness_%`	Score indicating the presence of live-performance elements, expressed on a 0–100 scale.
`speechiness_%`	Score indicating the amount of spoken-word content in the song, expressed on a 0–100 scale.

End of Assignment